

CARGO: A Mobile Manipulator Solution for Container Unloading

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Abstract—Shipping container unloading is a physically demanding task often carried out under challenging conditions, which motivates the use of automation. However, automating this process is complex due to the unpredictable sizes and quantities of each shipment. Existing solutions tend to be task-specific, rely on closed software stacks, and offer limited information on performance in non-controlled environments, which restricts their adaptability. We present *CARGO*, a modular pipeline that enables a mobile manipulator equipped with regular sensors and actuators to unload containers autonomously. The pipeline employs a predefined, layered workflow composed of reconfigurable modules that can be adapted to various robots, ensuring that all boxes in a stack are systematically handled. In simulation, the pipeline successfully unloaded a full container without collisions, thereby validating the complete workflow. Laboratory tests further confirmed these results, with the mobile manipulator successfully unloading boxes across multiple trials, with a success rate of 97 %. These results demonstrate that a versatile mobile manipulator can handle mixed box sizes and chaotic layouts using a generic, modular pipeline, highlighting a promising direction for flexible container-unloading automation.

Index Terms—Container unloading, mobile manipulator, modular pipeline, industrial robotics, logistics automation

I. INTRODUCTION

Global logistics networks rely heavily on shipping containers to move products efficiently across continents. Inside these containers, goods are typically tightly packed in cardboard boxes to maximize available space, resulting in dense and often irregular stacks [1]. When the containers arrive at distribution centers, unloading them becomes a demanding and repetitive task that requires reaching, bending, and lifting boxes under uncomfortable conditions. Workers must operate in confined spaces with limited lighting and extreme temperatures, often during overnight shifts. These factors, combined with the physical strain and labor shortage, have intensified the need for reliable robotic automation in the intralogistics sector.

Although some commercial systems demonstrate autonomous container unloading, they remain largely task-specific and rely on purpose-built hardware or software pipelines. Their software stacks are closed, and the available

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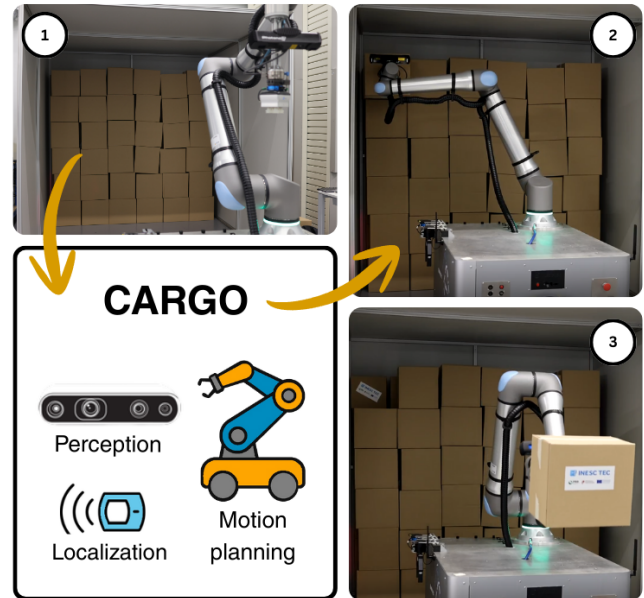


Fig. 1. CARGO pipeline running on a MoMa. 1) MoMa in a container unloading scenario. 2) MoMa detected the object and starts the picking sequence. 3) MoMa placing the box.

documentation offers only limited insight into performance, particularly when operating outside controlled conditions [2], [3]. In practice, industry increasingly demands flexible, reconfigurable robots that can be moved between different stages of a logistics workflow [4]. Our approach addresses this gap by emphasizing modularity and generality, enabling a single software pipeline to operate on general-purpose mobile manipulators (MoMa) with widely available sensors and actuators. Subsequently, our work is not only a solution for container unloading but also a step toward general-purpose mobile manipulation aligned with current research trends in real-world applications.

Building upon the hardware platform [5] successfully deployed in other [6], [7], this work presents the *CARGO* pipeline which applies a structured workflow to a MoMa equipped with off-the-shelf components. The pipeline consists of reconfigurable modules (encompassing perception, motion planning, manipulation, and other necessary components) that

can be reconfigured to the specific system in which they are deployed (Fig. 1). The requirements to use this pipeline are a vacuum gripper, a robotic arm, an RGB-D camera, and a holonomic drive platform (with omnidirectional wheels) supported by 2D laser scanners.

The main contributions of this work are: (i) A modular container-unloading pipeline designed for MoMa, requiring no task-specific hardware or parameter tuning across mixed box sizes; (ii) A robust perception and localization pipeline that maintains grasp reliability in low-light conditions and preserves localization accuracy inside the container despite degraded 2D scan geometry; (iii) A validated real-world deployment showing that a general-purpose MoMa can autonomously unload boxes of varying sizes under realistic operational constraints.

The remainder of this paper is organized as follows. Section II reviews relevant work on MoMa used for container unloading. Section III provides an overview of the container unloading task and the MoMa used in this work. Section IV details the methodology and the modules developed for the proposed pipeline. Section V presents the experimental evaluation in both simulation and laboratory conditions. Section VI concludes the paper and discusses future work.

II. STATE-OF-THE-ART

Research on autonomous container unloading includes several fields, such as perception, grasping, manipulation, motion planning, and task optimization. Over the years, researchers have approached the problem from different angles. One line of work focuses on sequence and plan optimization, aiming to determine efficient box-picking orders within dense stacks. Auth *et al.* [8] propose an A*-based unloading sequence planner that reduces the travel distance within containers. In contrast, Zhu *et al.* [9] attacked container unloading with a two-stage search algorithm that first identifies feasible box-removal sequences and then selects the one that minimizes handling effort. While these methods improve planning efficiency, they address only box ordering and do not consider the full system perspective.

A second line of research addresses full-system solutions. Stoyanov *et al.* [10] present a prototype robotic system that utilizes a conveyor-style mechanism to extract boxes from the container. Islam *et al.* [11] use a similar hardware setup but enhance the decision-making pipeline. However, both systems rely on custom hardware, limiting adaptability to other manipulators or tasks.

More recently, Hettiarachchi *et al.* [12] demonstrated a MoMa proof-of-concept for container loading and unloading. However, the system was only shown in simulation and relied on simplified perception and structured box layouts. Commercial systems, such as Boston Dynamics' Stretch [2] and Dexterity's Mech [3], further illustrate industrial interest, but they rely on specialized end-effectors and closed software stacks, which restricts their deployment flexibility.

All in all, the current literature indicates that container-unloading automation is feasible, but it still lacks a versatile,

hardware-agnostic pipeline capable of operating on MoMa under realistic container conditions, precisely the gap our system addresses.

III. SYSTEM OVERVIEW

The proposed pipeline enables a MoMa to autonomously unload containers in unknown or partially known environments. It coordinates perception, decision-making, and motion planning to safely handle boxes of varying sizes and positions. The following sections describe the task scenario and workflow architecture.

A. Task Description

The test scenario involves boxes with side lengths ranging from 20 cm to 60 cm, with multiple box sizes potentially present in the same shipment, and with a maximum weight of 20 kg. The number and dimensions of boxes are unknown prior to execution. Known parameters include the container dimensions and its initial orientation, with the opening facing the robot.

Considering the container reference frame, with the x -axis denoting depth, the y -axis denoting width, and the z -axis denoting height, we define a layer as the set of boxes lying on the same yz plane at a given depth x . In essence, each layer corresponds to a visible wall of boxes. Each layer is then subdivided into sections, which are smaller spatial regions based on the camera's Field of View (FoV). The system processes these sections sequentially, allowing task execution to adapt in real time as perception feedback updates the scene.

B. Workflow Architecture

The CARGO pipeline employs a workflow on the MoMa composed of individualized, reconfigurable modules that we refer to as *skills* (explained in detail in Sec. IV). These skills are created using ROS¹ actions, which makes them easy to reuse and adapt across different systems. Fig. 2 shows a simplified view of the workflow structure.

The system hierarchically organizes tasks. Initially, 2D laser scan data is used to identify the next layer of boxes. Then, that wall of boxes is segmented into sections based on the camera's FoV. The robot's base positions itself within reachable range of the robotic arm for each section, allowing the perception system to identify a target box. The manipulator then executes pick-and-place operations (picking the box and placing it in the desired final pose, which may be onto the MoMa itself or on top of another automated machine or conveyor) until the section is cleared. The process continues to the next section, and once all sections are completed, it advances to the subsequent layer. Using this structured layered approach we ensure efficient workspace coverage while maintaining both perception and manipulation feasibility.

¹<https://www.ros.org/>

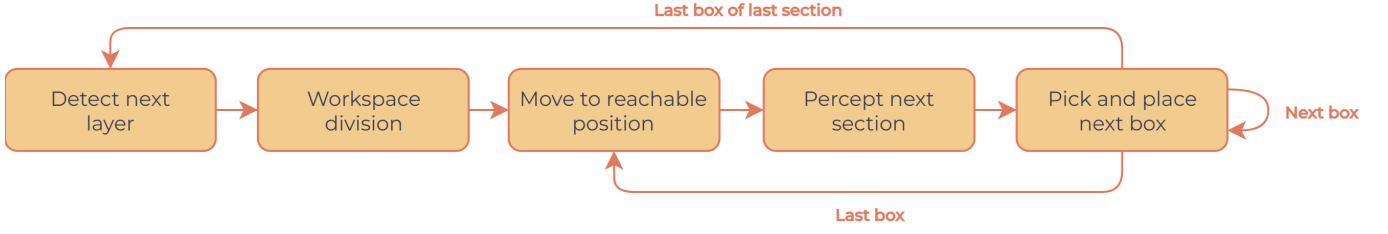


Fig. 2. System workflow overview

C. Robotic system

The experimental hardware utilized in this study is detailed in [5]. It features a custom in-house developed mobile base with mecanum wheels, enabling holonomic movement and SICK S300 lasers for localization and security purposes. The manipulator is a Universal Robots UR20 robotic arm with an OnRobot VGC10 gripper attached to its tool flange, along with a Photoneo MotionCAM-3D-S+ RGB-D camera for perception. This system has previously succeeded in naval, intralogistics, and aerospace applications [6], [7].

IV. DEVELOPED SKILLS

This section presents the main skills of the autonomous container unloading pipeline. All skills are parameterized with runtime-configurable inputs, making them easily adaptable to different robots and applications.

A. Next Layer Detection

At the start of the task, the robot faces an open container, but its precise location is unknown. To estimate where the next layer of boxes begins, the system first relies on the 2D laser scan data. A configurable number of scans, N , is collected to reduce noise and filter out outliers, forming the set $\mathcal{S}$. From each scan, only the points that fall within a chosen angular window Δ around a target direction ϕ are kept. In our setup, ϕ corresponds to the robot's forward direction, since the container is assumed to be located straight ahead. The filtered points form the set $\mathcal{P}$, as defined in Eq. 1, where r_k and θ_k denote the polar coordinates of the k -th element of the 2D laser scan.

$$\mathcal{P} = \bigcup_{i=1}^N \left\{ (r_k, \theta_k) \in \mathcal{S}_i \mid \theta_k \in \left[\phi \pm \frac{\Delta}{2} \right], k \in \mathbb{Z} \right\} \quad (1)$$

The points in $\mathcal{P}$ are subsequently grouped along the horizontal axis *with respect to* (w.r.t.) the robot frame. The cluster that contains the highest number of points is chosen as the next layer of boxes, and a linear regression is performed to estimate the location of that layer, $\mathcal{L}$. An example of the algorithm's execution is illustrated in Fig. 3.

B. Workspace Division

Due to the limited FoV of the end-effector camera, the robot cannot capture all boxes in a single scan. To ensure complete coverage, the container workspace is divided into overlapping sections, which the robot processes sequentially.

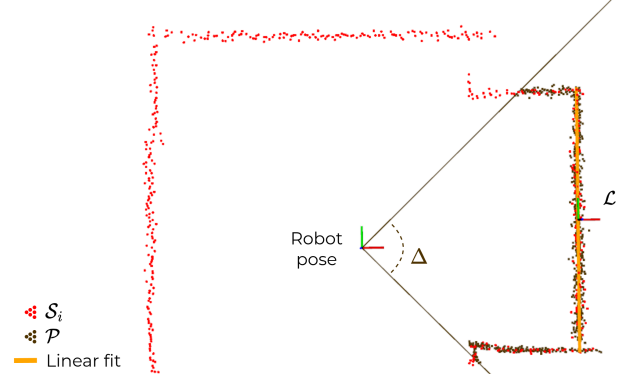


Fig. 3. Estimation of layer in the 2D laser scan data.

The division of each layer into sections is influenced by the dimensions of the container workspace (w_s), the scanning area at a given distance based on the camera's FoV (c_{fov}), and a configurable overlap (o) to ensure that even boxes that are only partially visible are fully captured in at least one section. The number of sections along each axis is calculated as outlined in Eq. 2, with $\mathcal{L}_f$ representing the last detected layer frame.

$$N_{splits,i} = \left\lceil \frac{w_{s,i} - o}{c_{fov,i} - o} \right\rceil, i \in \{y, z\} \text{ w.r.t } \mathcal{L}_f \quad (2)$$

C. Move Within Reach

Determining the optimal placement of the robot's base required analyzing the arm's reachability as a function of distance from its base. We sampled multiple joint configurations within the hardware limits, computing the end-effector position and manipulability of each configuration using the Yoshikawa index [13]. Each reachable pose was then associated with its manipulability, prioritizing configurations that offered higher dexterity.

The resulting data was aggregated into a histogram of radius values, which was normalized and smoothed using a cubic spline to produce a continuous reachability profile. This profile was interpreted as a probability distribution, allowing us to compute a confidence interval for reliable operation. In practice, we defined the 90% confidence range as the feasible zone for mobile base placement. Fig. 4 shows an example of the resulting reachability curve for the arm.

Using this reachability analysis, the system selects the optimal mobile base position for each section of a container

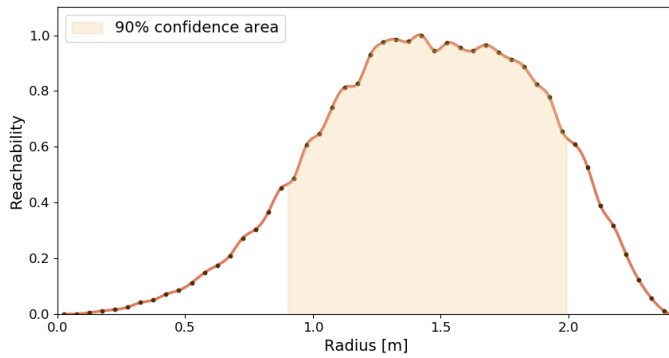


Fig. 4. Reachability analysis showing the confidence interval of end-effector reachability based on manipulability-weighted sampling.

layer. The base is positioned to horizontally center the arm relative to the current section, ensuring that the farthest boxes remain within the upper limit of the confidence range, while the closest boxes do not violate the lower limit.

Once the target base pose is defined, the robot executes a straight-line motion from its current location. This motion relies on predefined primitives stored in a dictionary, which encode repeatable transitions between task-specific robot states, ensuring consistent and safe execution.

For the custom mobile base, localization is handled differently depending on the environment. Inside the warehouse, where the environment is known, the perfect match algorithm [14] provides accurate positioning. When the robot reaches the front of the container (an unknown environment), localization switches to a scan-matching tracker. This tracker registers the latest 2D laser scan against a continuously updated local map. Ray tracing with a hysteresis window prevents nearby points from merging unnecessarily, thereby avoiding the integration of sensor noise into the local map, while an inliers-based policy decides when to incorporate new scans into the map. This strategy maintains accurate localization within the container, even as environment changes would provoke differences between the observed scan and the reference map in traditionally static map approaches [15].

D. Perception and Grasping

After acquiring a scan, the perception pipeline from [6], configured to detect box-shaped objects, is applied. The sensor provides RGB-D data by employing a structured-light method. First, the RGB image is processed using a hybrid approach that employs a YOLOv11 neural network [16] and analytical segmentation/filtering methods to perform 2D segmentation, resulting in 2D object masks. Objects appearing on the image borders are excluded, as they are likely only partially visible. The section-based scanning strategy ensures that such boxes will be fully observed in subsequent scans.

Each mask is projected onto the 3D point cloud through a back-projection step that uses the camera's intrinsic parameters, isolating the region of interest associated with each detected object. Several filtering stages then remove invalid

detections, such as boxes located outside the current layer or beyond acceptable geometric bounds.

A set of heuristics defines a consistent picking order, prioritizing boxes on top before those underneath and following a left-to-right sweep to maintain stable handling. After these steps, the pipeline outputs the full 6D pose of the next object to grasp.

Afterwards, multiple grasp poses are generated, and the grasp selection pipeline from [7] is then applied to select the best one. The evaluation utilizes heuristics to determine which grasp poses are feasible, taking into account arm joint limits, object reachability, and spatial constraints imposed by the container walls and ceiling. The output is a grasp pose that maximizes the likelihood of a successful pick, which is then passed to the motion planner for execution.

Our method accommodates various box sizes and ensures reliable detection even in low-light conditions, thanks to the camera's exposure control and built-in illumination. As a result, the perception stage remains robust throughout the container-unloading process.

E. Picking-and-Placing

Reliable and collision-free arm motion inside a container requires a strategy that works within tight geometric constraints. This work employs a heuristic motion scheme built around predefined motion primitives. These primitives, stored in a dictionary similar to the one used for the mobile base, use the structured nature of typical container packing to manipulate boxes safely without requiring online collision checking.

When a box is detected, the arm moves to a pre-grasp pose offset from the approach direction. It then executes a slow, linear motion to the grasp point to ensure precise alignment. After grasping, the box is lifted slightly to reduce friction with the box below (Fig. 5a), and then pulled straight out (Fig. 5b) to accommodate the full range of expected box depths, given that only a front-facing view is available. The exact depth is unknown at the time of grasp. If the box is close to a container side wall, an additional lateral shift repositions it toward the container center (Fig. 5c), reducing the likelihood of collisions during rotation. Finally, the arm rotates and transports the box to a predefined drop location behind the robot (Fig. 5d-f).

Additionally, another heuristic was applied to adjust the wrist orientation based on the vertical position of the target box. For boxes near the top of the container, the wrist maintains a natural upward orientation (Fig. 6.1). For boxes near or on the floor, the wrist is inverted, allowing the arm to reach lower positions without colliding with the floor (Fig. 6.2). This adaptation extends the effective workspace while ensuring safe execution across the full container height.

V. EXPERIMENTAL EVALUATION

The proposed framework was evaluated through a two-stage experimental process. First, we validated the system in simulation to ensure safety and verify the workflow under controlled conditions. We then progressed to lab experiments in

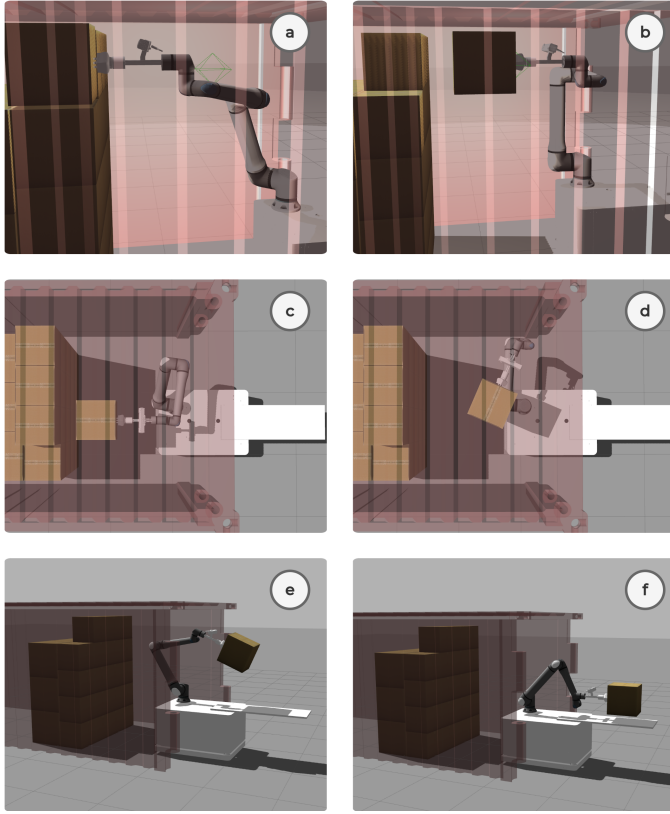


Fig. 5. Demonstration of arm movement heuristic in simulation.

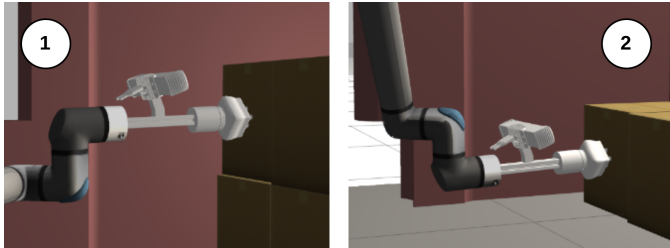


Fig. 6. Wrist orientation heuristics: 1) top-layer box with wrist up. 2) bottom-layer box with wrist inverted.

a container-like mock-up. A video demonstrating the system's behavior is available at this link².

Tab. I summarizes the results, comparing simulation and laboratory performance. These findings demonstrate that the proposed framework can autonomously unload containers with mixed box sizes using only off-the-shelf MoMa components while maintaining safety, efficiency, and reliability in realistic logistics conditions.

A. Simulation testing

Simulation experiments were conducted using the Gazebo simulator³ due to its native ROS integration. The simulated environment included a 20-foot container with 170 boxes

TABLE I
COMPARISON OF SIMULATION AND LABORATORY RESULTS FOR
CONTAINER UNLOADING EXPERIMENTS.

Metric	Simulation	Laboratory
Total boxes tested	170	288
Successful picks	170	279
Failures	0	9
- Gripper suction	n.a	4
- Perception	n.a	3
- Box tilt/stack interaction	0	2
Collisions with environment	0	0
Average execution time per box	30 s	20 s
Layer unload time (36 boxes)	35 min	20 min
Base repositioning per layer	1	1

*n.a - not applicable

distributed across four layers. Box dimensions ranged from 20 cm to 60 cm, covering the expected predefined range. The simulation focused on validating the workflow, motion planning, and execution pipelines prior to hardware testing.

Perception was bypassed in the simulation to isolate the motion and manipulation workflow under controlled conditions. The ground-truth box positions were provided to the grasping skill, while boxes were still assigned to sections according to their location to maintain workflow validation. Grasping was simulated through the implementation of a custom Gazebo plugin. When the gripper approaches a box within a predefined threshold, a virtual joint is established, effectively securing the box to the gripper. It is essential to note that factors such as vacuum efficiency and the weight of the box were not considered in this simulation.

Arm motion was executed using *MoveIt⁴ Pilz*, simulating *MoveJ* (linear joint-space movement) and *MoveL* (linear cartesian-space movement) commands consistent with real industrial manipulators. Mobile base motion was simulated using the *Planar Move Plugin³*, which enables planar displacement without detailed mecanum wheel dynamics.

Across multiple trials, the robot successfully removed all boxes without collisions or failed grasps. Boxes of all sizes and layer configurations were handled effectively, demonstrating that the pipeline generalizes to mixed workloads. The workflow and motion primitives in simulation accurately reflect the real system, providing confidence that hardware experiments would yield similar performance.

The timing difference between simulation and real-world execution comes from the motion planning process. In the simulation, *MoveIt* needs more time to generate arm trajectories, while the actual robotic system can execute these motions almost instantly.

²<https://www.youtube.com/watch?v=VDqVGoDK2Ng>

³<https://classic.gazebo.org/>

⁴<https://moveit.ai/>

B. Laboratory validation

Following simulation validation, the system was evaluated in a laboratory setup that replicated a real container environment (as illustrated in Fig. 1). Multiple single-layer trials were conducted. Three different box sizes were used. However, each layer contained only one size, randomly arranged to create an irregular/disorganized stack. All boxes were empty. The vacuum gripper used has a limited payload, particularly for laterally oriented grasps, which constrains the maximum weight the system can handle. In this phase, the full perception and manipulation pipeline are the focus of the testing experiments.

The robot required a single base placement per layer, with no repositioning. Across all trials, 279 out of 288 boxes were successfully picked and placed, which results in a success rate of 97%. For the 9 failures that occurred: 4 were due to incomplete suction by the gripper, 3 due to the perception pipeline (either failing to detect a box or incorrectly filtering it out), and 2 caused by tilted boxes, where removing one box displaced another in the stack. No collisions with container walls, ceiling, or floor were observed. The perception failures consistently occurred in the same region, suggesting a potential systematic error.

Average execution time per box was approximately 20 s (excluding perception computation), resulting in roughly 20 min to unload a layer of 36 boxes of 35 cm each. So, if the container was full of boxes of this size, it would take less than 6 h to unload. Current velocities were limited to half of the arm's maximum to ensure stability of the mobile base. Despite these conservative speeds, the system maintained accurate grasping, robust perception under varying lighting conditions, and reliable localization within the container.

VI. CONCLUSIONS AND FUTURE WORK

We introduced CARGO, a pipeline that enables a MoMa to autonomously unload containers. The system integrates reconfigurable modules, making it applicable across different MoMa hardware. Its modular structure allows each component to operate independently while maintaining cohesive, safe, and efficient manipulation, handling mixed box sizes, irregular layouts, and varying environmental conditions.

We evaluated the workflow in both simulation and laboratory conditions. The simulation results showed that the system can unload a full container without collisions, thereby validating the complete end-to-end workflow. The experiments also revealed natural limitations of using a general-purpose robot, particularly when unloading layers composed of many small boxes (e.g. 20 cm). Because the robot can only lift one box at a time, the unloading time increases substantially for dense layouts of small items.

The laboratory tests confirmed that the same workflow can be executed consistently on the real platform, highlighting the benefits of a reconfigurable architecture for adaptation and incremental refinement. These trials also exposed additional constraints, such as the limited camera FoV, which increases the number of perception cycles required per layer, and the lack of global awareness, which may cause local instabilities

in the stack when removing a single box. Furthermore, the payload limitations of the gripper combined with lateral grasps restrict the system from reaching the originally targeted 20 kg, making the current setup more suitable for lighter materials.

Considering the achieved results of 97% success, future work will focus on reducing the observed failures by integrating a vacuum monitoring module to ensure a box is successfully picked or detect if it falls from the gripper, and implementing fail-safe mechanisms to prevent a grasp from destabilizing the surrounding stack. Additionally, other efforts will include improving trajectory generation to remove constraints tied to specific box sizes, testing the framework with other MoMa systems, and enhancing the overall robustness of the system for real-world industrial deployment.

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